Xinglang Zhang

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Education

- **Huazhong University of Science and Technology (HUST)** *Sept 2024 – Jun 2028*
  B.S. in Cyber Science and Engineering
  - **GPA:** 4.80/5.0 **Weighted Avg:** 94.51 **Rank:** **1/108**
  - **Languages:** English (TOEFL 5.5/6; W6 R6 L5.5 S5; CET-4: 609; CET-6: 611), Chinese (Native)
  - **Research Interests:** LLM Reasoning, LLM Test-Time Training, Neural-Symbolic AI, Agents, Tool Learning

Selected Honors and Awards

- **Outstanding Student**, Qiming College *(Highest Honor for HUST Undergraduates; Top 1.85%)* 2025 – 2027
- **Jin Yugang Lan-Ying Scholarship** *(Top 1%)* 2025 – 2026
- **National Scholarship** *(Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide)* 2025 – 2026
- **Triple-A ("Sanhao") Student Scholarship** *(Top 5.5%)* 2025 – 2026
- **National Scholarship** *(Highest Honor for Chinese Undergraduates; Top 0.3% Nationwide)* 2024 – 2025
- **Triple-A ("Sanhao") Student Scholarship** *(Top 5.5%)* 2024 – 2025

Publications

* Equal contribution.

[1] **Xinglang Zhang***, Yunyao Zhang*, et al. **Logical Phase Transitions: Understanding Collapse in LLM Logical Reasoning**. **ACL 2026 Main Conference**. [Code]

[2] Yunyao Zhang, **Xinglang Zhang**, et al. **Semantic-Aware Logical Reasoning via a Semiotic Framework**. **ACL 2026, Oral**. [Code]

[3] **Xinglang Zhang**, Cheng Qian. **TRACE: Transfer-Aware Test-Time Training for Tool-Using LLMs**. **Conference Submission**.

[4] **Xinglang Zhang**, Yuanmeng Xiang, et al. **Too Good to Be Real? Diagnosing and Reducing the Gap Between AI Preference and Real User Engagement**. **Conference Submission**.

[5] Yunyao Zhang*, **Xinglang Zhang***, et al. **Semiotic Logical Hexagon Theory for LLM Logical Reasoning**. **Journal Submission**. [Code]

[6] Zikai Song, Xiajie Li, Yunyao Zhang, **Xinglang Zhang**, et al. **Social Intelligence Modeling: A Comprehensive Survey from Social Perception to Social Simulation**. [Code]

[7] Yunyao Zhang, Zuocheng Ying, **Xinglang Zhang**. **IntervenSim: Intervention-Aware Social Network Simulation for Opinion Dynamics**. [Code]

[8] Zhiqing Cui, Xinxiang Yin, Yihong Tang, **Xinglang Zhang**, et al. **EarthVerse: Benchmarking Scientific Agents Across Dynamic Earth Systems and Natural Hazards**. [Project]

[9] Xinxiang Yin, **Xinglang Zhang**, et al. **From Reflection to Co-Evolution: Self-Evolving Agents with Locally Verified Memory**.

Selected Funded Projects

- **HUST Undergraduate Natural Science Fund, 2026** — *Trustworthy Enhancement for Complex Reasoning*; Project Lead; **CNY 50,000 (approx. US$7,450)**
- **College Student Innovation and Entrepreneurship Program (University Level)** — *LLM Logical Reasoning through Philosophical Paradigms*; Project Lead; **CNY 4,000 (approx. US$600)**
- **College Student Innovation and Entrepreneurship Program (Hubei Provincial Level)** — *Trustworthy Enhancement of LLMs via a Neuro-Symbolic Framework*; Project Lead; **CNY 20,000 (approx. US$2,980)**

Research Experience

- **Understanding Multi-Agent Systems** *Nov 2026 – May 2027*
  Research Intern, System Intelligence Group, *Microsoft Research Asia* Beijing, China
  Research Focus: Multi-Agent Systems

- **TRACE: Transfer-Aware Test-Time Training** *Jul 2026 – Present*
  Research Intern, *BLENDER Lab, UIUC*
  Mentor: Cheng Qian **Lab Director:** Heng Ji

  - **Framework:** Defined **Transfer Value** as the target-specific, state-dependent benefit of learning from a self-generated experience.
  - **Method:** Developed **TRACE**, a gold-free curriculum combining Target Relevance, Current-State Transferability, and Marginal Complementarity through temporary updates and probes.
  - **Findings:** Current-state feedback improved selection over stale signals; dynamic selection consistently outperformed a fixed curriculum.
  - **Results:** Across four tool-use benchmarks and four main models, improved Overall Exact accuracy over frozen models by **13.10–24.42 percentage points**. First-author **ACL 2027** submission [3].

- **Anchor: Locally Verified Memory for Self-Evolving Agents** *Aug 2026 – Oct 2026*
  Research Collaborator, *HKU CDS*
  Advisor: Jian Qian

  - **Framework.** Co-proposed Anchor, a self-evolving agent memory framework that evaluates candidate memories through paired decision replay, combining behavioral shift magnitude and source-role alignment to selectively retain useful experiences.
  - **Results.** Co-conducted experiments across ALFWorld, WebShop, and two LLM backbones. Achieved **96.27% ALFWorld success** and **73.50 WebShop score** with Qwen3.5-27B, outperforming the strongest baselines while retaining **41% fewer memories** than unfiltered storage. Second-author paper [9].

- **HexLogicAgent: Semantic Reasoning for Reliable LLMs** *Mar 2026 – Jul 2026*
  Research Assistant, Intelligent Media Computing and Network Security Lab, *HUST*
  Advisor: Zikai Song

  - **Framework:** Proposed **HexLogicAgent**, a semantic-aware logical reasoning framework that organizes proposition meanings through a logical hexagon before deduction and verification.
  - **Theory:** Developed a formal logical hexagon theory for semantic reasoning.
  - **Results:** Achieved consistent gains across five logical reasoning benchmarks.
  - **Outcome:** Co-first-author **Artificial Intelligence** submission [5].

- **OMRA: Synthetic–Real Preference Alignment** *Jan 2026 – May 2026*
  Project Leader, Intelligent Media Computing and Network Security Lab, *HUST*
  Advisor: Zikai Song

  - **Framework:** Proposed *Ontological Preference Measurement* to diagnose synthetic-real preference mismatch across logic, affect, and expression dimensions.
  - **Method:** Developed **OMRA** to mask and reconstruct reasoning scaffolds while preserving stance and coherence.
  - **Results:** Reduced marginal, geometric, and transfer gaps by **54.4% on average** across four LLM families; first-author **NAACL 2027** submission [4]. All three reviewers gave an **OA score of 3.5/5**; the Area Chair gave a **meta-review score of 4**; **top 8%** of submissions.

- **Logical Phase Transitions in LLM Reasoning** *Aug 2025 – Jan 2026*

  Project Leader, Intelligent Media Computing and Network Security Lab, HUST

  Advisor: Zikai Song

  - **Framework:** Identified **Logical Phase Transitions** in LLM reasoning and proposed a *Neural-Symbolic Tuning* framework to stabilize reasoning under increasing logical complexity.
  - **Results:** Across five benchmarks, the method mitigates reasoning collapse, yielding **+1.26 (naive)** and **+3.95 (CoT)** average accuracy gains with improved generalization.
  - **Outcome:** Published as first author in **ACL 2026 Main Conference [1]**.

- **LogicAgent: Semantic-Aware Logical Reasoning** *Feb 2025 – Aug 2025*

  Research Assistant, Intelligent Media Computing and Network Security Lab, HUST

  Advisor: Zikai Song

  - **Framework & Benchmark:** Proposed **LogicAgent**, a semiotic-aware logical reasoning framework, and introduced *RepublicQA*, a benchmark jointly modeling logical and semantic complexity.
  - **Results:** Achieved state-of-the-art results on RepublicQA (+**6.25%**) and strong generalization across four logical reasoning benchmarks (+**7.05%**).
  - **Outcome:** Co-authored paper published in **ACL 2026 Main Conference [2]**.

Leadership Experience

Class Representative *Sept 2024 – Present*

Qiming Experimental Class 2401

Organized academic events, managed administrative tasks, and mentored students to enhance class cohesion.

Head of Organization Department *Sept 2025 – Present*

School of Cyber Science and Engineering

Managed administrative and organizational affairs for the Communist Youth League.

Head of Lesson Planning *Apr 2025 – Present*

Peer Lecturer Group

Fostered academic excellence across the college through peer mentoring and curriculum planning.

Skills

ML & Research Stack

Python, C/C++, PyTorch, HuggingFace, LoRA, SFT, RLHF, vLLM

Mathematical Foundations

Linear Algebra, Probability Theory, Calculus, Discrete Mathematics

Academic Activities

Poster Presenter and Oral Presenter, ACL 2026 *July 1 – 8, 2026*

Selected Rising Star Poster Presenter, X-AGI Conference *Oct. 17 – 18, 2026*

Service

Reviewer, ACL ARR *January, May, and August 2026*